The ionization and/or excitation of atoms and molecules when the energies of nuclear particles are absorbed in matter is the basis for the detection of individual particles. Macroscopic collective effects, such as chemical changes and heat evolution, can also be used. The most important of the latter have been described before because of their importance for dose measurements (e.g. the blackening of photographic films and other chemical reactions, excitation of crystals (thermoluminescence), and heat evolved in calorimeters; Ch. 8). Although animals have no known senses for detection of nuclear radiation, it has been found that subletal but large radiation fields can affect animals in various ways such as disturbing the sleep of dogs or causing ants to follow a new pathway to avoid a hidden radiation source. Apollo astronauts observed scintillations in their eyes when their space ship crossed very intense showers of high energy cosmic rays. People who have been involved in criticality accidents experiencing high intensities of n and γ have noted a fluorescence in their eyes and felt a heat shock in their body.

However, we are not physiologically aware of the normal radiation fields of our environment. In such low fields we must entirely rely on instruments.

In this chapter we consider only the common techniques used for detection and quantitative measurement of *individual* nuclear particles. We also discuss the problem of proper preparation of the sample to be measured as well as consideration of the statistics of the counting of nuclear particles necessary to ensure proper *precision* (i.e. how well a value is determined) and *accuracy* (i.e. agreement between measured and true value).

9.1. TRACK MEASUREMENTS

The most striking evidence for the existence of atoms comes from the observation of tracks formed by nuclear particles in cloud chambers, in solids and in photographic emulsions. The tracks reveal individual nuclear reactions and radioactive decay processes. From a detailed study of such tracks, the mass, charge and energy of the particle can be determined.

The tracks formed can be directly observed by the naked eye in cloud and bubble chambers, but the tracks remain only for a short time before they fade. For a permanent record we must use photography. On the other hand, in solid state nuclear track detectors (SSNTD), of which the photographic emulsion is the most common variant, the tracks have a much longer lifetime during which they can be made permanent and visible by a suitable chemical treatment. Because of the much higher density of the absorber, the tracks are also much shorter and often therefore not visible for the naked eye. Thus the microscope is an essential tool for studying tracks in solids.

9.1.1. Cloud and bubble chambers

The principle of a *cloud chamber* is shown in Figure 9.1. A volume of saturated vapour contained in a vessel is made supersaturated through a sudden adiabatic expansion. When ionizing radiation passes through such a supersaturated vapour the ionization produced

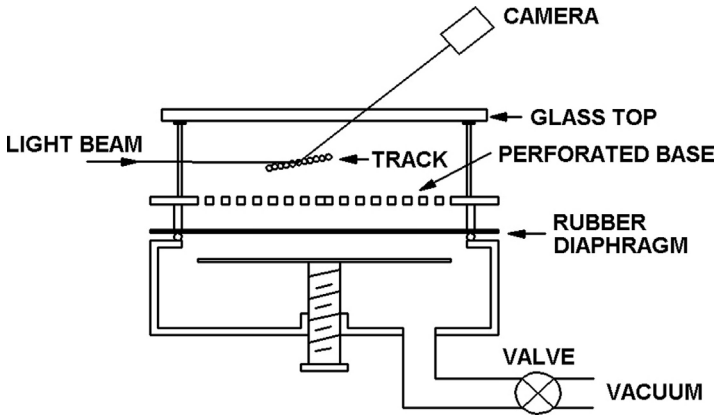


Figure 9.1 Principle of a cloud chamber.

in the vapour serves as condensation nuclei. As a result small droplets of liquid can be observed along the path of the radiation. These condensation tracks have a lifetime of less than a second and can be photographed through the chamber window. The density of the condensation depends on the ionization power of the projectile as well as on the nature of the vapour, which is often an alcohol or water. Cloud chamber photographs are shown in Figure 7.5.

In a similar fashion *bubble chambers* operate with superheated liquids in which gas bubbles are produced upon the passage of ionizing radiation. The most commonly used liquid in bubble chambers is hydrogen, and, as a consequence, the chamber must be operated at low temperatures (23 K for H_2). Since the liquid medium in a bubble chamber is much denser than the vapour medium in a cloud chamber, the former are more suitable for studies of reactions of more energetic projectiles. The high energy p-p reaction shown in Figure 2.2 has been recorded in a large hydrogen bubble chamber at CERN; see also Figure 7.18.

9.1.2. Solid state nuclear track detectors (SSNTD)

The main types of SSNTD (or DTD, for dielectric track detector) are photographic emulsions, crystals, glasses, and plastics. Because the density of these materials is much higher than for the gases often used in the previous detector group (section 9.1.1), nuclear particles can spend all their kinetic energy in these detectors, allowing identification of the particle. Since the SSNTD retains the particle path, they can be used to record reactions over a long time period. These advantages have made SSNTD especially valuable in the fields of cosmic ray physics, radiochemistry, and earth sciences.

Nuclear emulsions are similar to optical photographic emulsions. They contain AgBr crystals embedded in gelatin to which small amounts of sensitizing agents have been

added. The AgBr content is as much as four times (i.e. 80% AgBr) greater than in optical film. Also the crystals are much smaller (developed grain 0.1—0.6 μm) and well separated. The emulsions come in thicknesses from a few μm up to 1 mm. Nuclear radiation passing through the emulsion causes ionization and excitation which activates the AgBr crystals, producing a latent image of the particle path. Upon development the activated crystals serve as centers for further reduction to silver metal, leading to visible grains. It is assumed that at least 3 silver atoms must be activated to produce a visible grain, while about 30 atoms are needed for normal blackening. Each activated grain seems to require an average absorption of about 2.5 eV.

While the memory effect of the properly developed film is almost infinite, this is not the case for the latent image which slowly fades, depending on the number of originally activated silver atoms, the film type and external conditions like temperature, humidity, etc. When stored under ambient conditions, about 80% of the latent image disappears in half a year.

The developed grains form an interrupted track along the original path of the energetic particle (Fig. 9.2(a)). The specific energy loss of the particle, dE/dx (i.e. the stopping power of the absorber), depends on the mass, charge, and velocity of the particle, and on the composition of the absorber. From the track length, grain density (i.e. grains per track length), and gap length between the grains, the particle and its energy can be determined (cf. sections 7.3 and 8.1). For a given particle, the range is proportional to the energy as shown in Table 9.1. The range decreases with increasing mass of the particle and density of the absorber. The grain density depends on the specific ionization of the particle which does not vary linearly with the particle energy (or velocity), as seen from Figure 7.7; thus the grain density changes along the track.

Other solid material may be used as SSNTD instead of AgBr emulsions: *plastics* (cellulose nitrate and polycarbonate films), *glass*, *crystals*, etc. In order to make the tracks visible in the microscope the surface of the SSNTD must be polished and etched, usually with alkali.

Because of the natural radiation background, every SSNTD has a memory of past nuclear events, which must be erased as far as possible before a new exposure. In nuclear emulsions an α-radiation background of 20—60 tracks cm^{-2} per day is normal. The technique of *background eradication* prior to exposure may consist of treating an emulsion with chromic acid, H_2O_2-vapour or heating (annealing) a glass plate. Because this technique more easily removes weak images, it may also be used after exposure, e.g. to remove fainter α-tracks from heavier fission tracks.

Let us consider some examples of uses of SSNTD. Tracks obtained under various conditions are shown in Figure 9.2.

(i) As mentioned SSNTD has been used in cosmic ray experiments at high altitudes and in space journeys where the memory effect and simple construction make them especially suitable. Many elementary particles have been discovered by this

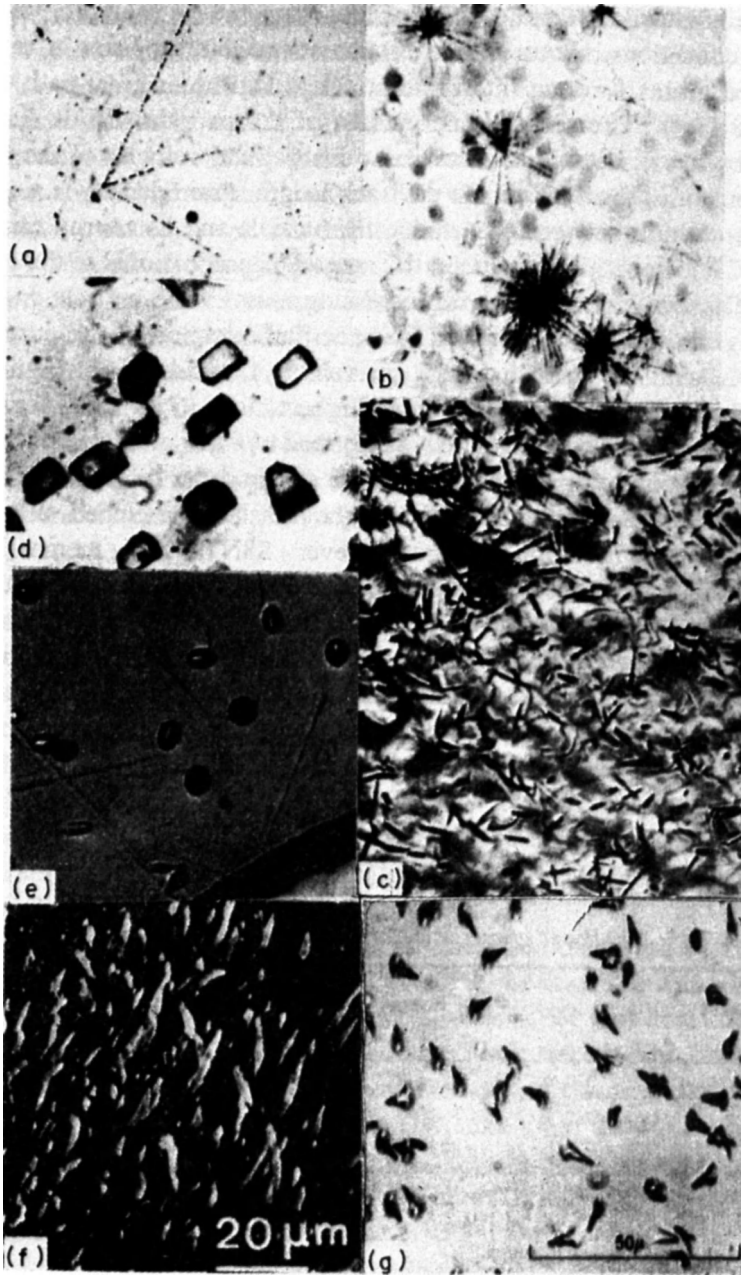


Figure 9.2 Tracks of highly ionizing particles in solid absorbers. (a) Alpha-tracks originating in the same point in a nuclear emulsion. (Acc. to P. Cuer.) (b) Autoradiography of a lung showing deposits of inhaled plutonium. (Acc. to B. A. Muggenburg.) (c) Fission tracks in neutron irradiated apatite containing some evenly distributed uranium. (Acc. to E. I. Hamilton.) (d) Neutron-induced fission tracks in muscovite mineral. (Acc. to E. I. Hamilton.) (e) Neutron-induced fission tracks in volcanic glass recovered from deep sea sediments. (Acc. to J. D. Macdogall.) (f) Fission tracks in mineral from the Oklo mine. (Acc. to J. C. Dran *et al.*) (g) Fission tracks from ^{252}Cf in Lexan polycarbonate. *(From Fleischer, Price, and Walker).*

Table 9.1 Range of energetic high-ionizing particles in various solids

Particle	Energy (MeV)	Absorber (Density)	Range (μm)
H	10	Ilford C2 (3.8)[a]	540
^{3}H	10	Ilford C2 (3.8)[a]	230
^{4}He	10	Ilford C2 (3.8)[a]	57
^{4}He (^{214}Bi)	7.7	Eastman NTA (3.6)	38
		Mica (3.1)	36
		Glass (2.5)	41
		Water (1.0)	60
^{4}He (^{238}U)	4.2	Mica (3.1)	13
^{4}He (U-series)		Pitchblende (7.0)	23[b]
		Carnotite (4.1)	32[b]
^{235}U ⌈ Light fiss. fragm.	~150	Eastman NTC (~3.4)	14 ⌉ ~25
⌊ Heavy fiss. fragm.		Eastman NTC (~3.4)	10.5 ⌋
^{238}U Both fiss. fragm.	~160	Leopoldite (~4)	20

[a]Density of AgBr 6.47; of gelatin 1.31.
[b]Range of the predominating α-particles.

technique, notably the π- and μ-mesons. Figure 13.2 shows tracks of high energy cosmic ray particles, probably iron atoms, which have been stopped in Apollo astronaut helmets.

(ii) Nuclear reactions can be studied by SSNTD. The target material may either be regular atoms of the detector (H, O, Ag, Si, etc.) or material introduced into the matrix, e.g. thin threads of target metals or uranium atoms. The former have been used in high energy physics for hadron-induced reactions, and the latter for studying fission processes. From experiments with uranium the frequency of spontaneous fission of ^{238}U has been determined, and also the rate of ternary fission and emission of long range α's.

(iii) When emulsions are dipped into solution, some of the dissolved material is soaked up or absorbed in the emulsion. For example, if the solution contained samarium, some α-tracks of its spontaneous decay (decay rate 127 Bq g^{-1}) appear in the emulsion (cf. Fig. 9.2(a)). Since ^{147}Sm has an isotopic abundance of 15%, its half-life is calculated to be 1.06×10^{11} y. The lower limit of detection is about 500 tracks cm^2 d^{-1}, so quite low decay rates can be accurately measured, making this a valuable technique for determination of long half-lives.

(iv) ^{222}Rn is released through the earth's surface from uranium minerals. The amount released varies not only with the uranium content and mineral type, but also with the time of the day; variation from 1−75 Bq l^{-1} has been registered during a 24 h period. To avoid this variation, cups containing a piece of plastic TD are placed upside down in shallow (0.5−1.0 m) holes for about 3 weeks, after which the SSNTD are etched and α-tracks from radon counted. Mineral bodies hundreds of

meters underground can be mapped with this technique in great detail in a reasonably short time. The US Geological Survey uses a similar technique to predict earthquakes; it has been observed that just before earthquakes the radon concentration first increases, then suddenly decreases, the minimum being observed about one week before the earthquake appears.

(v) The average radon concentration in houses can be measured by hanging a plastic film inside the house over a time period of some weeks. The film is returned to a laboratory where it is etched and the number of α-tracks per unit area counted.

(vi) Fission fragments make dense tracks in all solid material. The tracks are short and thick: in a crystal material like zircon (a common mineral, $ZrSiO_4$) they may not be more than 10^{-2} μm (10 nm) in diameter, and 10−20 μm in length. They are therefore not visible even in the best optical microscopes. Using scanning electron microscopy, it has been found that the hole formed retains the crystal structure or regains it (Fig. 9.2(d)). On the other hand, if the track is formed in glass, a gas bubble appears instead of a track (Fig. 9.2(e)); these slightly elongated bubbles can be distinguished from other completely spherical bubbles formed by other processes. To make the tracks visible, the specimen is embedded in a resin, then one surface is ground and carefully polished after which it is dipped in an acid, e.g. HF. Because of defects in the crystal structure along the fission track, the track and its close surroundings are attacked by the acid, and the diameter of the track increases a hundredfold to a micron or so. The tracks are then visible under a microscope with a magnification of 500−1000×.

This procedure has been used as an analytical tool for determination of uranium and plutonium in geological and environmental samples. In this technique, the sample (either a ground and polished surface of a mineral, or a dust sample on tape) is firmly pressed against a photographic film, and the package is irradiated by slow neutrons. From the fission track count of the developed film the uranium or plutonium content can be calculated. Thus a Swedish shale was found to contain 4 ± 1 ppm U, and a bottom sediment in a Nagasaki water reservoir 0.44 ± 0.04 Bq ^{239}Pu per kg sediment. In the latter case, the ratio between the number of fission tracks N_{ft} and α-tracks $N_{\alpha t}$ from ^{239}Pu is

$$N_{ft}/N_{\alpha t} = \sigma_f \varphi/\lambda_\alpha \tag{9.1}$$

where σ_f is the fission cross section, φ the n-flux, and λ_α the decay constant of ^{239}Pu. This technique is very useful for routine measurements of fissionable material in very low concentrations and for the safeguards program. Figure 9.2(c) shows fission tracks in a uranium-containing mineral which has been exposed to neutrons.

Fission track counting is also important for dating of geological samples (Ch. 13) and for estimation of the maximum temperature experienced by sedimentary rocks. Temperature history information can be obtained from a combination of age (estimated by

other radioactive methods, cf. Ch. 13), uranium content of crystals of several minerals in the rock, their fission track count and track-length distribution caused by thermal annealing. Zircon, titanite and apatite are examples of three such minerals in order of increasing annealing temperature. The temperature history of a rock formation is important in oil prospecting operations as the maximum temperature seems to be a useful indicator on whether to expect oil, natural gas or nothing. For too low temperatures neither oil nor gas is expected, for intermediate temperatures oil may be present and for high temperatures only natural gas.

9.2. GENERAL PROPERTIES OF DETECTORS

The principle for detection of nuclear radiation is the same as discussed previously for both the interaction of radiation with matter and radiation production namely: excitation, ionisation, deflection, and atom recoil. In the case of detection only the first two of the interactions are primarily used even if, naturally, the others also occur. A nuclear particle entering a detector produces excitation and ionization, both of which can be used for detection. When the excitation is followed by fluorescent de-excitation (section 7.8.4 and Ch. 8) the light emitted can be registered by light-sensitive devices, e.g. the photo-multiplier tube (PMT) which transforms the light into an electric current. Scintillation and Čerenkov detectors are based on light emission. A similar current is generated when production of *charge carriers* (i.e. ions, electrons and holes) takes place between the charged electrodes of a detector. Detectors based on the production of charge carriers are either gas-filled (ion chambers, proportional and Geiger-Müller tubes in which charge carriers are produced by ionization of a gas) or solid, usually semiconductor crystals. In the latter case electrons and holes are produced in pairs (section 9.4).

An ionizing particle or photon will produce a collectable charge ΔQ in the detector

$$\Delta Q = 1.60 \times 10^{-19} E_{\text{loss}} \eta w^{-1} \tag{9.2}$$

where E_{loss} is the total energy lost by the particle to the detector, η is the charge collection efficiency, w is the energy required for the formation of a pair of charge carriers in the detector medium, while the constant is the charge (Coulomb) of a single charge carrier (the pairs must be regarded as singles because they move in opposite direction in the electric field due to their opposite charges).

Because of the short duration of the absorption process for a single particle (10^{-4}—10^{-9} s) the short current, i, is referred to as a *pulse* of charge ΔQ

$$i = \Delta Q / \Delta t \tag{9.3}$$

If this current passes through a resistor R it will produce a voltage pulse (cf. Eqn (9.2))

$$\Delta V = R \, \Delta Q / \Delta t \tag{9.4}$$